DS501: Data Science - Project #4: Predictive Modeling Competition

Topic: Applying Supervised Machine Learning Models for Prediction and Model Selection
Due Date: [Date 20 days from release]
Total Points: 300 points (means this project has three times weights of previous projects)

1. Project Goal (The Challenge)

You are tasked with participating in a simulated predictive modeling competition. Your objective is to build the most accurate machine learning model to predict **housing prices (a regression task)** and **loan default risk (a classification task)** using a real-world dataset.

The core requirement of this project is to implement and compare performance across the major model families introduced in Lecture #10:

1. **Linear Models:** Linear Regression (using the Normal Equation or Gradient Descent).
2. **Tree-Based Models:** Decision Tree, Random Forest.
3. **Instance-Based Models:** K-Nearest Neighbors (KNN).
4. **Kernel Methods:** Support Vector Machine (SVM) (for classification task only).

Your final report must present a clear process for model training and output results clearly. In the code, provide a detailed explanation on the important code.

Dataset

You will use the **Ames Housing Dataset** (for Regression,use attached AmesHousing.csv attached in the assignment section of CANVAS) and the **Default of Credit Card Clients Dataset** (for Classification, use attached defaultofcreditcardclients.csv).

- *Note: For regression purposes, the predicted attribute (y) is "SalePrice". For classification purposes, the predict attribute (i.e., y) is "default_payment_next_month"*

3. Required Deliverables and Tasks

Task 1: Data Preprocessing and EDA (15 Points) This task has been implemented in the attached project4.ipynb, you can directly use it.

1. **Exploratory Data Analysis (EDA):** Perform basic EDA on both datasets. Identify data types, missing values, and potential outliers. Visualize key relationships between important features and the target variable.
2. **Preprocessing:** Implement necessary steps for cleaning and preparing the data for modeling:
    - Handle missing values (imputation).
    - Encode categorical features (one-hot encoding or similar).

- **Scaling/Normalization:** Apply feature scaling (e.g., standardization or MinMax scaling) and discuss *why* scaling is crucial for models like KNN and SVM, referencing the need to prevent distance measures from being dominated by high-variance attributes (as discussed on Slide 67).

2.

Task 2: Model Implementation and Training (250 Points)

For each task (Regression and Classification), you must implement and train the required models. All models must be implemented using scikit-learn.

Model Family	Regression Task (Housing Price)	Classification Task (Loan Default)
Linear	**Linear Regression** (Must implement using both **analytical solution** (e.g., numpy.linalg.solve or sklearn.linear_model.LinearRegression) and **Gradient Descent** (sklearn.linear_model.SGDRegressor))	**Linear Classification** (Use a linear model like LogisticRegression as a baseline)
Tree-Based	**Decision Tree Regressor**	**Decision Tree Classifier**
	Random Forest Regressor	**Random Forest Classifier**
Instance-Based	**K-Nearest Neighbors Regressor**	**K-Nearest Neighbors Classifier**
Kernel Method	N/A	**Support Vector Machine (SVM) Classifier** (SVC)

Task 3: Final Report and Analysis (50 Points)

Prepare a detailed analysis that includes the following sections:

1. **Results Table:** Present the final performance metrics for **all** trained models on the held-out test sets.
   - **Regression Metric:** Root Mean Squared Error (RMSE).
   - **Classification Metric:** F1-score and Accuracy.
2. **Model Comparison and Selection:**
   - Identify the best-performing model for the Regression task and the Classification task.

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Submission Guidelines:

1. A Jupyter Notebook (.ipynb) containing all code (Tasks 1-3).